02/17

Monday, 17 February 2025 9:35 AM

→ Stability of physics → PDE

→ But the discretization has its own stability → this is what we are concerned about.

Consistency	Convergence	Stability
As $\Delta x \to 0$, the solution looks similar to PDE	As $\Delta x \to 0$, the solution is the exact soln. of PDE	

↳ Can only prove for linear problems

↳ For non-linear can only linearize the check stability locally

↳ If the errors are bounded/ contained locally then don't care about the non-linearity

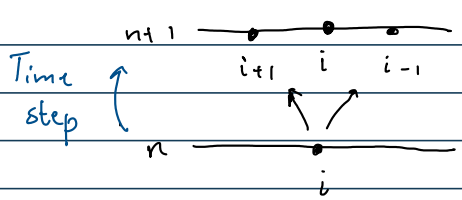
Checking the engineering:

1) Change to finite volume problem

(In - out

2) Transportive error - a disturbance is propagated (discrete perturbation analysis)

↳ error is felt beyond the solution



· Central differencing is adequate

3) Dissipation error - If the highest time derivative is one, then even order derivatives represent diffusion (momentum) or dissipation (mech. energy to thermal energy)

↳ Hard to avoid



G_{VW} → upp wind

↳ Shortest wavelength have higher dissipation

↳ Lax - Wendroff

Dispersive medium → waves w/ different wavelengths travel at different speeds in that medium

Non dispersive medium → all waves travel at the same speed

Wtf is a wave???

Dissipation ≠ Dispersion

↳ wher order is three